Chapter 7. Blocking and Confounding in the 2^k Factorial Design

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1 Blocking a Replicated Design

- If there are n replicates of the 2^k design, each replicate can be run as a **block** (e.g. one batch of raw material, one day).
- Replicates are run in block order; runs *within* a block are randomized.

Example 7.1: Chemical process in three blocks

Block 1	Block 2	Block 3
$(1) = 28$	$(1) = 25$	$(1) = 27$
$a = 36$	$a = 32$	$a = 32$
$b = 18$	$b = 19$	$b = 23$
$ab = 31$	$ab = 30$	$ab = 29$
$B_1 = 113$	$B_2 = 106$	$B_3 = 111$

Without blocks (Ch. 6): $SS_A = \frac{1}{2^2 \cdot 3}(ab+a-b-(1))^2 = 208.33$, $A = \frac{1}{2 \cdot 3}(ab+a-b-(1)) = 8.33$; error df $2^2(3-1) = 8$.
Treat the replicate as a block, exactly like the RCBD of Ch. 4, $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$ with the $2^2 = 4$ treatment combinations as "treatments". In Ch. 4, $SS_{\text{Block}} = \sum_i \sum_j (\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2 = \frac{1}{a}\sum_j y_{\cdot j}^2 - \frac{1}{N} y_{\cdot\cdot}^2$. Here each block has $2^2 = 4$ runs:

$$SS_{\text{Blocks}} = \frac{1}{4}\sum_{j=1}^{3} B_j^2 - \frac{1}{12} y_{\cdot\cdot}^2 = \frac{113^2 + 106^2 + 111^2}{4} - \frac{330^2}{12}$$

$$= \frac{36326}{4} - 9075 = 9081.5 - 9075 = 6.50, \qquad \text{df} = b - 1 = 2,$$
$$SS_E = 31.34 - 6.50 = 24.84, \qquad \text{df} = 8 - 2 = 6.$$

Source	SS	DF	MS	F_0
Blocks	6.50	2	3.25	
A	208.33	1	208.33	50.32
B	75.00	1	75.00	18.12
AB	8.33	1	8.33	2.01
Error	24.84	6	4.14	
Total	323.00	11		

Conclusions are unchanged (block effect is small); the effect estimates are identical because each block contains a full replicate.

2 Confounding in Blocks: the Unreplicated Case

When a full replicate cannot be run in one block (block size $< 2^k$), some effect information must be sacrificed: it becomes **confounded** (indistinguishable) with blocks. Usually a high-order interaction is chosen.

2.1 2^2 in two blocks

Block 1	Block 2
(1)	a
ab	b

Trt	I	A	B	AB	Block
(1)	+	−	−	+	1
a	+	+	−	−	2
b	+	−	+	−	2
ab	+	+	+	+	1

$A = \frac{1}{2}(ab + a - (1) - b)$ and $B = \frac{1}{2}(ab + b - (1) - a)$ are **unaffected by blocking**: each estimate has one plus and one minus treatment combination in each block. But $AB = \frac{1}{2}(ab + (1) - a - b) = \frac{1}{2}[(\text{block 1}) - (\text{block 2})]$ is **confounded with blocks**.

Formal argument [Supplement]. Add block effects δ_1, δ_2 to every observation in blocks 1 and 2. Then

$$\mathrm{E}(\text{Contrast}_A) = \mathrm{E}[ab] + \mathrm{E}[a] - \mathrm{E}[(1)] - \mathrm{E}[b] \ \ni \ \delta_1 + \delta_2 - \delta_1 - \delta_2 = 0,$$
$$\mathrm{E}(\text{Contrast}_{AB}) = \mathrm{E}[ab] + \mathrm{E}[(1)] - \mathrm{E}[a] - \mathrm{E}[b] \ \ni \ \delta_1 + \delta_1 - \delta_2 - \delta_2 = 2(\delta_1 - \delta_2) \neq 0.$$

Thus the A and B contrasts are free of block effects, while the AB contrast estimates AB-interaction + block difference: they cannot be separated. ■

2.2 2^3 in two blocks with ABC confounded

Trt	I	A	B	AB	C	AC	BC	ABC	Block
(1)	+	−	−	+	−	+	+	−	1
a	+	+	−	−	−	−	+	+	2
b	+	−	+	−	−	+	−	+	2
ab	+	+	+	+	−	−	−	−	1
c	+	−	−	+	+	−	−	+	2
ac	+	+	−	−	+	+	−	−	1
bc	+	−	+	−	+	−	+	−	1
abc	+	+	+	+	+	+	+	+	2

Block 1 $= \{(1), ab, ac, bc\}$ (the $-$ signs of ABC), Block 2 $= \{a, b, c, abc\}$ (the $+$ signs). Every other column has two $+$ and two $-$ within each block (orthogonality), so all effects except ABC are estimated free of blocks.

3 Constructing the Blocks: the Defining Contrast

Definition 7.2: Defining contrast

$$L = \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_k x_k,$$

$x_i \in \{0, 1\}$ is the level of the ith factor in a treatment combination, and $\alpha_i \in \{0, 1\}$ is the exponent of the ith factor in the effect to be confounded. Treatment combinations with the same $L \pmod 2$ go to the same block.

For ABC: $\alpha_1 = \alpha_2 = \alpha_3 = 1$, $L = x_1 + x_2 + x_3$.

Trt	L	mod 2
(1)	$0+0+0=0$	0
a	$1+0+0=1$	1
b	$0+1+0=1$	1
ab	$1+1+0=2$	0
c	$0+0+1=1$	1
ac	$1+0+1=2$	0
bc	$0+1+1=2$	0
abc	$1+1+1=3$	1

Block 1 ($L = 0$): $(1), ab, ac, bc$; Block 2 ($L = 1$): a, b, c, abc.

Remark 7.3: Why L works

The sign of a treatment combination in the column of effect $A^{\alpha_1}B^{\alpha_2}C^{\alpha_3}$ is $\prod_i (2x_i - 1)^{\alpha_i} = (-1)^{\sum_i \alpha_i (1 - x_i)}$, which depends only on $\sum \alpha_i x_i \pmod 2$ (for fixed α). So $L \bmod 2$ splits the runs exactly by the sign of the confounded column.

3.1 The principal block and group structure

- The **principal block** is the block containing (1).

- The principal block is a *group* under multiplication mod 2 (exponents mod 2): any element except (1) can be generated by multiplying two other elements of the principal block. For ABC confounded:

$$ab \cdot bc = ab^2c = ac, \qquad ab \cdot ac = a^2bc = bc, \qquad bc \cdot ac = abc^2 = ab.$$

- The other block is a coset: multiply one element of the new block by every element of the principal block. Since b is in the other block:

$$b \cdot (1) = b, \quad b \cdot ab = ab^2 = a, \quad b \cdot ac = abc, \quad b \cdot bc = b^2c = c.$$

Closure [Supplement]. If $L(u) \equiv 0$ and $L(v) \equiv 0$, then for the product uv (exponents added mod 2) $L(uv) \equiv L(u) + L(v) \equiv 0$, because L is linear mod 2. Similarly if $L(w) \equiv 1$ then $L(wu) \equiv 1$ for every u in the principal block. ■

3.2 More blocks (four blocks)

To run a 2^k in 2^p blocks choose p independent effects to confound; their **generalized interactions** (products mod 2) are automatically confounded too. E.g. a 2^3 in 4 blocks confounding AB and AC also confounds $AB \cdot AC = A^2BC = BC$: $3 = 2^2 - 1$ effects are lost (one per block df). Blocks are formed by the pairs $(L_1, L_2) = (x_1 + x_2,\ x_1 + x_3)$ mod 2: $(0,0) : \{(1), abc\}$, $(1,1) : \{a, bc\}$, $(1,0) : \{b, ac\}$, $(0,1) : \{c, ab\}$.

4 Block Confounding with Replication

4.1 Complete confounding

The same effect is confounded in every replicate. 2^3 in two blocks, three replicates, ABC confounded in all:

Rep 1		Rep 2		Rep 3	
Bl 1	Bl 2	Bl 1	Bl 2	Bl 1	Bl 2
(1)	a	ac	a	c	(1)
ab	b	(1)	c	abc	ac
ac	c	ab	abc	b	bc
bc	abc	bc	b	a	ab

(Block labels within a replicate are arbitrary; each replicate splits $\{(1), ab, ac, bc\}$ vs $\{a, b, c, abc\}$, and the assignment of runs within blocks is randomized.)

Source	DF
Replications	2
Blocks (or ABC)	1
Replication × blocks	2
A, B, AB, C, AC, BC	1 each (6)
Error (replication × all other effects)	12
Total	23

The 6 block df ($= 2 \cdot 3$ blocks $-1 = 5$) split as 2 (replicates) +1 (ABC/blocks) +2 (rep × blocks); the error df $= 3 \cdot 6 - 6 = 12$ are the replicate × effect interactions for the six estimable effects. ABC cannot be tested.

4.2 Partial confounding

Confound a *different* interaction in each replicate, so information on every interaction is retained from the replicates where it is not confounded. 2^3, four replicates, confounding ABC, AB, AC, BC in replicates 1–4:

Rep 1 (ABC)		Rep 2 (AB)		Rep 3 (AC)		Rep 4 (BC)	
Bl 1	Bl 2	Bl 1	Bl 2	Bl 1	Bl 2	Bl 1	Bl 2
(1)	a	(1)	a	(1)	a	(1)	b
ab	b	c	b	b	c	a	c
ac	c	ab	ac	ac	ab	bc	ab
bc	abc	abc	bc	abc	bc	abc	ac

(Check rep 2 with $L = x_1 + x_2$: $(1), c, ab, abc$ have $L \equiv 0$.)

Source	DF
Replications	3
Blocks within replications	4
ABC (rep 1), AB (rep 2), AC (rep 3), BC (rep 4)	1 each
A, B, C	1 each
AB (from reps 1,3,4), AC (reps 1,2,4), BC (reps 1,2,3), ABC (reps 2,3,4)	1 each
Error	17
Total	31

DF check: $31 = 3 + 4 + 7 + 17$. Each partially confounded interaction is estimated from the 3 replicates in which it is unconfounded, so its SS is $(\text{Contrast}^{(3\text{ reps})})^2/(3 \cdot 2^3)$ and it has relative information 3/4.

Example 7.4: ABC in rep 1, AB in rep 2, BC in rep 3

(a) *Blocks.* Rep 1 ($L = x_1 + x_2 + x_3$): $\{(1), ab, ac, bc\} \,|\, \{a, b, c, abc\}$. Rep 2 ($L = x_1 + x_2$): $\{(1), c, ab, abc\} \,|\, \{a, b, ac, bc\}$. Rep 3 ($L = x_2 + x_3$): $\{(1), a, bc, abc\} \,|\, \{b, c, ab, ac\}$.
(b) *ANOVA skeleton* ($N = 24$):

Source	DF
Replications	2
Blocks within replications (ABC rep 1, AB rep 2, BC rep 3)	3
A, B, C, AC (never confounded)	1 each (4)
AB (reps 1, 3), BC (reps 1, 2), ABC (reps 2, 3)	1 each (3)
Error	11
Total	23

Error df $= 23 - 2 - 3 - 7 = 11$.

5 Selected exercises

Motivation. The exercises verify that the confounded effect and the block contrast are literally the same sum of squares, prove the counting rule for 2^p blocks stated above, and show that confounding does not remove block noise but concentrates it in one chosen effect.

Exercise 7.1 (Confounded effect = blocks; Montgomery 7.22)

(a) For the 2^2 design in two blocks with AB confounded, prove algebraically that $SS_{AB} = SS_{\text{Blocks}}$.
(b) More generally, for an unreplicated 2^k in two blocks confounding effect E, show $SS_E = SS_{\text{Blocks}}$.

Solution.

(b) contains (a) ($k = 2$, $E = AB$, blocks $\{(1), ab\}$ and $\{a, b\}$).

Step 1. Let B_+ and B_- be the totals of the two blocks, the runs with sign $+$ and $-$ in column E, each of 2^{k-1} runs. Then $\text{Contrast}_E = B_+ - B_-$ and $SS_E = (B_+ - B_-)^2/2^k$.

Step 2. As for blocks in Ch. 4, with $y_{..} = B_+ + B_-$,

$$SS_{\text{Blocks}} = \frac{B_+^2 + B_-^2}{2^{k-1}} - \frac{(B_+ + B_-)^2}{2^k} = \frac{2B_+^2 + 2B_-^2 - (B_+ + B_-)^2}{2^k} = \frac{(B_+ - B_-)^2}{2^k}.$$

For $k = 2$: $SS_{AB} = \frac{1}{4}\{ab + (1) - a - b\}^2 = SS_{\text{Blocks}}$. The two sources are one and the same single degree of freedom, so the ANOVA can list only one of them. ■

Exercise 7.2 (The algebra of 2^p blocks)

Identify an effect $A^{\alpha_1} \cdots K^{\alpha_k}$ with $\alpha \in \{0,1\}^k \setminus \{\mathbf{0}\}$ and a run with $\mathbf{x} \in \{0,1\}^k$. Blocks are formed by p linearly independent (over arithmetic mod 2) defining contrasts $\alpha^{(1)}, \ldots, \alpha^{(p)}$: runs with equal $(L_1(\mathbf{x}), \ldots, L_p(\mathbf{x}))$, $L_r(\mathbf{x}) = \alpha^{(r)} \cdot \mathbf{x} \bmod 2$, share a block.
(a) Show that there are 2^p blocks of 2^{k-p} runs each.
(b) Show that an effect α is constant in sign within every block iff α is a nonzero sum (mod 2) of some of the $\alpha^{(r)}$ (a generalized interaction), so exactly $2^p - 1$ effects are confounded.
(c) Show that every other effect has equally many $+$ and $-$ runs in every block, so its estimate is free of additive block effects.

Solution [Supplement]. Work in the vector space $\mathbb{F}_2^k$ (arithmetic mod 2). By the remark "Why L works", the sign of run $\mathbf{x}$ in column α is $\pm(-1)^{\alpha \cdot \mathbf{x}}$ with the sign depending only on α, so it is determined by $\alpha \cdot \mathbf{x} \bmod 2$.

(a) The map $T(\mathbf{x}) = (L_1(\mathbf{x}), \ldots, L_p(\mathbf{x}))$ is linear $\mathbb{F}_2^k \to \mathbb{F}_2^p$ of rank p (its rows are independent), hence onto, and each fibre $T^{-1}(\mathbf{b})$ is a coset of $\ker T$, which has 2^{k-p} elements. The blocks are these 2^p fibres.

(b)

Step 1 ($\Leftarrow$). If $\alpha = \sum_{r \in R} \alpha^{(r)}$, then $\alpha \cdot \mathbf{x} = \sum_{r \in R} L_r(\mathbf{x})$, which is constant on each block.

Step 2 ($\Rightarrow$). Suppose $\alpha \cdot \mathbf{x}$ is constant on every block. On the principal block $\ker T$ it equals its value at $\mathbf{0}$, namely 0, so $\alpha \in (\ker T)^\perp$. Since $\dim \ker T = k - p$, $\dim (\ker T)^\perp = p$, and the row space of T is contained in $(\ker T)^\perp$ and also has dimension p, so the two coincide: α is a combination of the $\alpha^{(r)}$.

Step 3 (count). The row space has 2^p elements; removing $\mathbf{0}$ leaves $2^p - 1$ confounded effects, one per block degree of freedom.

(c) If α is not in the row space, then by Step 2 the linear functional $\mathbf{x} \mapsto \alpha \cdot \mathbf{x}$ is not identically 0 on $\ker T$. A nonzero linear functional on $\ker T$ takes the values 0 and 1 equally often (2^{k-p-1} times each, its kernel being a subspace of index 2). On a block $\mathbf{x}_0 + \ker T$, $\alpha \cdot \mathbf{x} = \alpha \cdot \mathbf{x}_0 + \alpha \cdot \mathbf{z}$, so the same balance holds. Hence the block effects cancel in Contrast_α. ■

Exercise 7.3 (Where does the block noise go?)

An unreplicated 2^k ($N = 2^k$) must be run on two batches of $N/2$ runs, with random batch effects δ_1, δ_2 iid with variance σ_δ^2, independent of the errors (variance σ^2). Each effect is estimated by $\hat{E} = \text{Contrast}/2^{k-1}$.
(a) If the highest-order interaction is confounded with batches, show that every other effect has variance $\sigma^2/2^{k-2}$, while the confounded contrast picks up an extra $2\sigma_\delta^2$.
(b) If instead the runs are split into the two batches completely at random, show that every effect has average variance $\sigma^2/2^{k-2} + 2\sigma_\delta^2/(N-1)$.
(c) Compare the total extra variance summed over the $N-1$ effects in (a) and (b).

Solution [Supplement].

Step 1 (general form). With column $\mathbf{c}$ ($N/2$ entries $+1$, $N/2$ entries -1) and $s = \sum_{\mathbf{x}\in\text{batch }1} c(\mathbf{x})$, the batch part of the contrast is $\delta_1 s + \delta_2(0-s) = (\delta_1 - \delta_2)s$. The errors contribute $\sigma^2 \mathbf{c}^T\mathbf{c} = N\sigma^2$ to Var(Contrast) and are independent of the batches, so given the assignment

$$\text{Var}(\hat{E}) = \frac{N\sigma^2 + 2\sigma_\delta^2 s^2}{(N/2)^2} = \frac{\sigma^2}{2^{k-2}} + \frac{8\sigma_\delta^2 s^2}{N^2}.$$

(a) By Exercise 7.2(c) ($p = 1$), $s = 0$ for every unconfounded effect. For the confounded one $s = \pm N/2$, giving the extra $8\sigma_\delta^2 (N/2)^2/N^2 = 2\sigma_\delta^2$.

(b) Batch 1 is a random half of the runs, so the number H of $+1$ entries in it is hypergeometric (population N, $N/2$ successes, $N/2$ draws), with $\text{Var}H = \frac{N}{2}\cdot\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{N/2}{N-1} = \frac{N^2}{16(N-1)}$ and $\text{E}H = N/4$. Since $s = 2H - N/2$, $\text{E}s^2 = 4\text{Var}H = N^2/\{4(N-1)\}$, and the average extra variance is $8\sigma_\delta^2 \text{E}s^2/N^2 = 2\sigma_\delta^2/(N-1)$.

(c) Both totals equal $2\sigma_\delta^2$: in (b) it is spread evenly over all $N-1$ effects, in (a) it is concentrated in the one effect we chose to sacrifice, leaving all the others with the precision of a homogeneous experiment. ■